

Simplified DES:

S-DES, that is, Simplified DES, has similar properties and structure to DES with much smaller parameters.

Figure C.1 illustrates the overall structure of the simplified DES, which we will refer to as SDES.

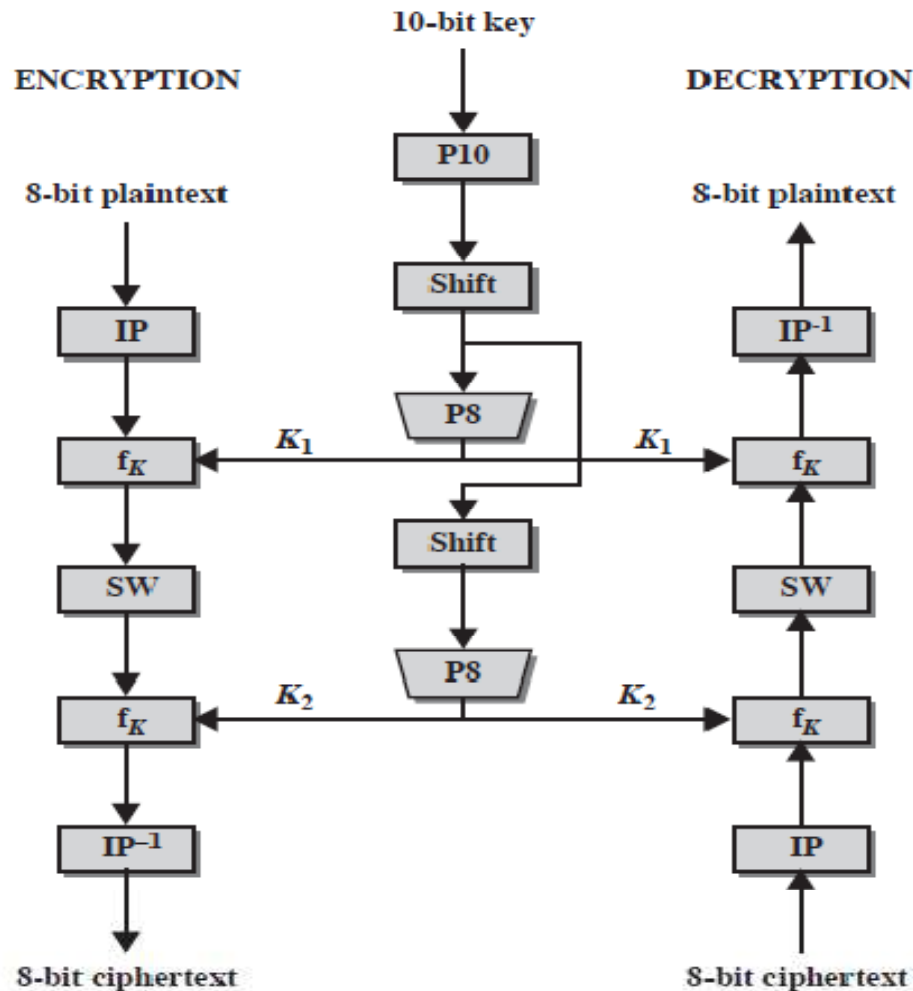


Figure C.1: Simplified DES Scheme

- ✓ The S-DES **encryption algorithm** takes
 - 8-bit block of plaintext
 - 10-bit key as input
 - Produces an 8-bit block of ciphertext as output.
- ✓ The S-DES **decryption algorithm** takes
 - 8-bit block of ciphertext
 - 10-bit key used to produce that ciphertext as input
 - Produces the original 8-bit block of plaintext.
- ✓ The encryption algorithm involves **five functions**:
 - An initial permutation (IP);

- A complex function labeled f_K , which involves both permutation and substitution operations and depends on a key input;
- A simple permutation function that switches (SW) the two halves of the data;
- The function f_K again;
- A permutation function that is the inverse of the initial permutation (IP^{-1}).
- We can concisely express the encryption algorithm as a composition of functions:

$$IP^{-1} \circ f_{K_2} \circ SW \circ f_{K_1} \circ IP$$

which can also be written as

$$\text{Ciphertext} = IP^{-1} (f_{K_2} (SW (f_{K_1} (IP(\text{plaintext}))))))$$

where

$$K_1 = P8(\text{Shift}(P10(\text{key})))$$

$$K_2 = P8(\text{Shift}(\text{Shift}(P10(\text{key}))))$$

- ✓ Decryption is the reverse of encryption as shown in **Figure C.1**:

$$\text{Plaintext} = IP^{-1} (f_{K_1} (SW (f_{K_2} (IP(\text{ciphertext}))))))$$

NOTE:

Definition: If f and g are two functions, then the function F with the equation $y = F(x) = g[f(x)]$ is called the **composition** of f and g and is denoted as $F = g \circ f$.

➤ **S-DES Key Generation:**

- S-DES depends on the use of a 10-bit key shared between sender and receiver.
- From this key, two 8-bit sub keys are produced for use in particular stages of the encryption and decryption algorithm as shown in **figure C.2**

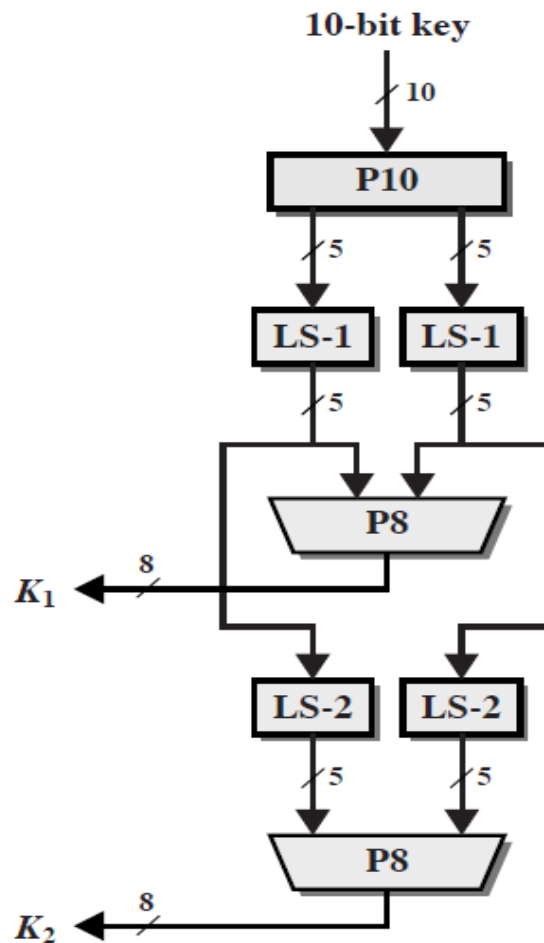


Figure C.2: Key Generation for SDES

- The 10-bit key is designated as $(k_1, k_2, k_3, k_4, k_5, k_6, k_7, k_8, k_9, k_{10})$.
- Then the permutation P10 is defined as:

$$P10(k_1, k_2, k_3, k_4, k_5, k_6, k_7, k_8, k_9, k_{10}) = (k_3, k_5, k_2, k_7, k_4, k_{10}, k_1, k_9, k_8, k_6)$$

P10 can be concisely defined by the display:

P10									
3	5	2	7	4	10	1	9	8	6

NOTE: Refer the example below this explanation to understand S-DES properly.

- Now perform a circular left shift (LS-1), or rotation, separately on the first five bits and the second five bits.
- Next apply P8, which picks out and permutes 8 of the 10 bits according to the following rule:

P8							
6	3	7	4	8	5	10	9

- The result is sub key 1 (**K1**).
- Further, on each of the 5-bit strings produced by the two LS-1 functions, a circular left shift of 2 bit positions is performed.
- Finally, P8 is applied again to produce sub key **K2**.

➤ **SDES Encryption:**

Encryption involves the sequential application of five functions as shown in **figure C.3**:

• **Initial Permutations:**

The input to the algorithm is 8-bit block of plaintext, which are first permuted using the IP function:

IP							
2	6	3	1	4	8	5	7

This retains all 8 bits of the plaintext but mixes them up.

• **The Function f_K**

The most complex component of S-DES is the function f_K , which consists of a combination of permutation and substitution functions.

The functions can be expressed as follows:

Let L and R be the leftmost 4 bits and rightmost 4 bits of the 8-bit input to f_K .

Let F be a mapping from 4-bit strings to 4-bit strings. Then,

$$f_K(L, R) = (L \oplus F(R, SK), R)$$

Where SK is a sub key and $\oplus$ is the bit-by-bit exclusive-OR function.

The mapping F can be described as follows:

- o The input is a 4-bit number, R, the rightmost 4 bits of the 8-bit input to f_K .
- o The first operation is an expansion/permutation operation on this R. After expansion, 8-bit number is produced from the 4-bit number.
- o The 8-bit sub key $K1 = (k_{11}, k_{12}, k_{13}, k_{14}, k_{15}, k_{16}, k_{17}, k_{18})$ is added to this value using exclusive-OR, producing new 8-bit output.
- o The first 4 bits are fed into the S-box S0 to produce a 2-bit output, and the remaining 4 bits are fed into S1 to produce another 2-bit output.
- o The S-boxes operate as follows. The first and fourth input bits are treated as a 2-bit number that specify a row of the S-box, and the second and third input bits specify a column of the S-box. The entry in that row and column, in base 2, is the 2-bit output.

The 4 bits produced by S0 and S1 undergo a further permutation as follows:

P4			
2	4	3	1

The output of P4 is the output of the function F.

• The Switch Function

The function f_K only alters the leftmost 4 bits of the input. The switch function (SW) interchanges the left and right 4 bits so that the second instance of f_K operates on a different 4 bits.

In this second instance, the E/P, S0, S1, and P4 functions are the same. The only difference is that the key input is $K2$.

• Final Permutations

At the end of the algorithm, the following inverse permutation is used on the 8-bits at hand:

IP ⁻¹							
4	1	3	5	7	2	8	6

NOTE: It is advisable to write P10, P8, IP and IP⁻¹ values as shown here. However, if you forget those values, do not write wrong ones. Just mention which operation is performed.

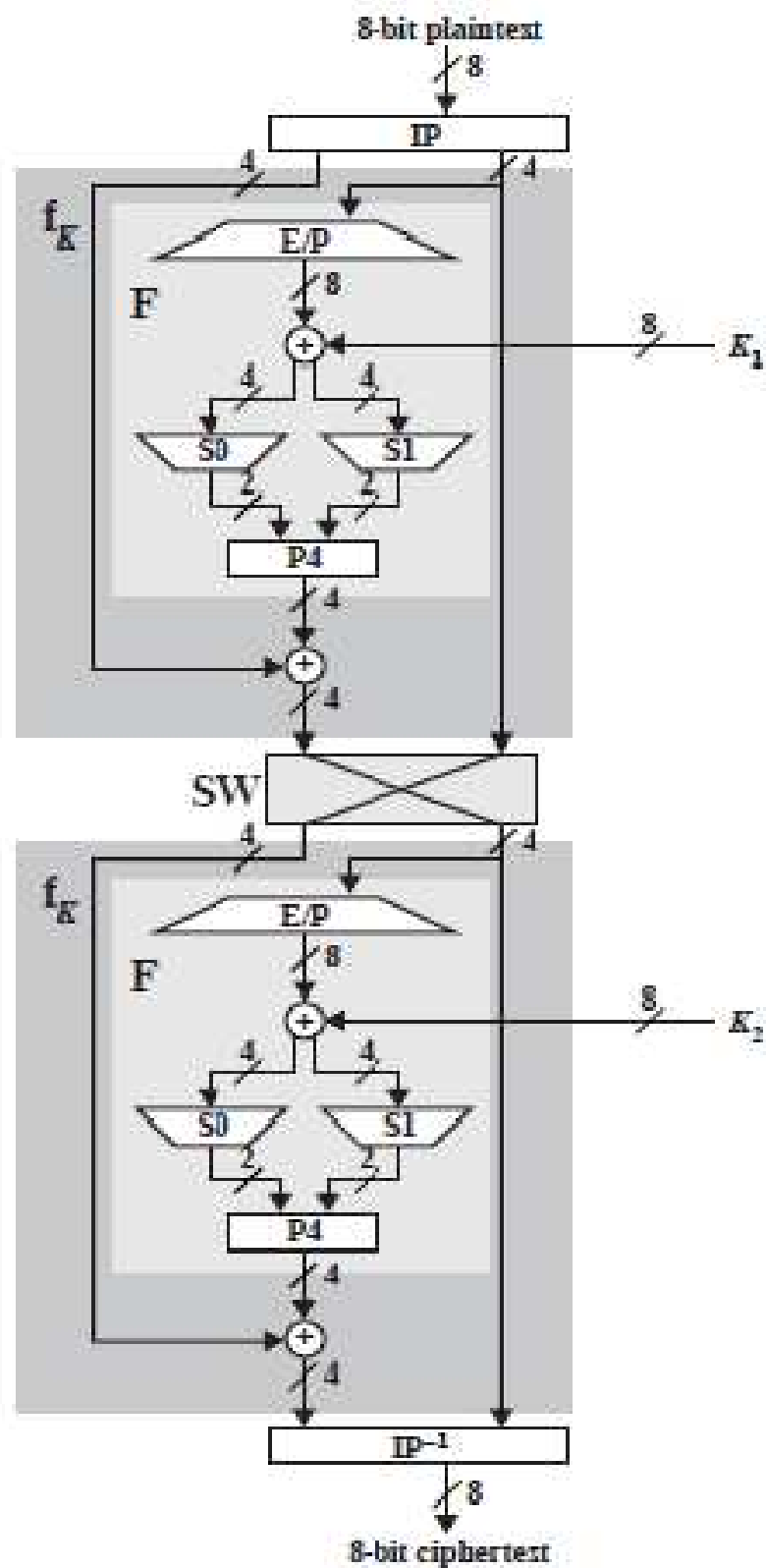


Figure: SDES Encryption details

NOTE:

E/P							
4	1	2	3	2	3	4	1

$$\begin{array}{c|c|c|c} n_4 & n_1 & n_2 & n_3 \\ n_2 & n_3 & n_4 & n_1 \end{array}$$

$$\begin{array}{c|c|c|c} n_4 \oplus k_{11} & n_1 \oplus k_{12} & n_2 \oplus k_{13} & n_3 \oplus k_{14} \\ n_2 \oplus k_{15} & n_3 \oplus k_{16} & n_4 \oplus k_{17} & n_1 \oplus k_{18} \end{array}$$

Renaming these 8 bits:

$$\begin{array}{c|c|c|c} P_{0,0} & P_{0,1} & P_{0,2} & P_{0,3} \\ P_{1,0} & P_{1,1} & P_{1,2} & P_{1,3} \end{array}$$

$P_{0,0}$, $P_{0,1}$, $P_{0,2}$ and $P_{0,3}$ are fed into the S-box S_0 to produce a 2-bit output,
The remaining 4 bits ($P_{1,0}$, $P_{1,1}$, $P_{1,2}$ and $P_{1,3}$) are fed into S_1 to produce another 2-bit output.

These two boxes are defined as follows:

$$S_0 = \begin{array}{c|c} & \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 3 & 2 \end{bmatrix} \end{array} \quad S_1 = \begin{array}{c|c} & \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{array}{c} 0 \\ 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix} \end{array}$$

The S-boxes operate as follows:

The first and fourth input bits are treated as a 2-bit number that specify a row of the S-box, and the second and third input bits specify a column of the S-box. The entry in that row and column, in base 2, is the 2-bit output.

EXAMPLE (Merely for better understanding of S-DES):

GIVEN:

Plain Text- 10111101

Key- 1010000010

METHOD:

Obtaining K1 & K2 from main key:

- 1) Applying P10 on key, we get: 1000001100
- 2) Applying Circular left shift by 1 bit on two 5 bits part, we get: 0000111000
- 3) Applying P8, we get **K1=10100100**
- 4) Applying Circular left shift by 2 bits on step no. 2, we get: 0010000011
- 5) Applying P8, we get **K2=01000011**

Now we have: PT=10111101

Key=1010000010

K1=10100100

K2=01000011

Steps for S-DES Encryption:

- 1) Initial permutation on plain text: IP(PT) = 01111110
- 2) Now applying fk1 on 0111 1110:

$$fk1(01111110) = [(0111) \text{ XOR } (F(1110, sk)), (1110)]$$

For the above, we need to calculate F(1110,sk) using K1:

- i) Apply E/P on 1110 we get: 0111 1101
- ii) XOR with K1: 11011001
- iii) Pass 1101 to S0 box and we get : 11
- iv) Pass 1001 to S1 box and we get : 10
- v) Combining both results from S-boxes we get : 1101
- vi) Then applying P4 on it we get : 1011

This is our required F, that is, $F(1110, sk) = 1101$

Now for:

$$\begin{aligned}fk1(01111110) &= [(0111) \text{ XOR } (F(1110, K1)), (1110)] \\&= [(0111) \text{ XOR } 1101), (1110)] \\&= [(1100), (1110)]\end{aligned}$$

Now we have as an intermediary output: 1100 1110

3) Applying SW we get: 1110 1100

4) Now applying fk_2 on 1110 1100:

$$fk_2(11101100) = [(1110) \text{ XOR } (F(1100, sk)), (1100)]$$

For the above we need to calculate $F(1100, sk)$ by K_2 :

i) Apply E/P on 1100 we get: 0110 1001

ii) XOR with K_2 : 0010 1010

iii) Pass 0010 to S_0 box and we get: 00

iv) Pass 1010 to S_1 box and we get: 00

v) Combining both results from boxes' we get : 0000

vi) Then Apply p_4 on it we get: 0000

This is our required F , that is $F(1100, sk) = 0000$

Now for:

$$fk_2(11101100) = [(1110) \text{ XOR } (F(1100, K_2)), (1100)]$$

$$= [(1110) \text{ XOR } 0000), (1100)]$$

$$= [(1110), (1100)]$$

Now we have at the end: 1110 1100

5) Apply IP^{-1} on it we get: 0111 0101 which is our Cipher Text

CT= 01110101